# Generative AI

- AI that **creates new content** (text, images, audio, video, code)
- Learns **patterns** from **data** → **generates** original **output**
- **Traditional AI** → **classifies/predicts** (e.g., spam detection)
- **Generative AI** → **creates new data** (e.g., write an email)




## Types

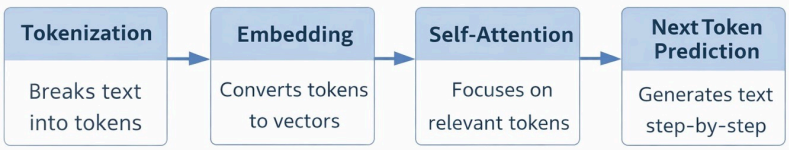
- **Text** → LLMs (e.g., **ChatGPT, Claude**)
- **Images** → Diffusion models (e.g., DALL-E, **Midjourney**)
- **Audio** → Tools like **ElevenLabs**, Suno
- **Video** → Tools like Sora, **Runway**

Claude

Gemini

## LLMs

- **Tokenize** input
- Use **self-attention** to understand context
- **Predict** next **token** step-by-step
- **Trained** on massive text **data**
- Intelligence emerges from scale




## How Diffusion Models Work ?

- Learn to **reverse noise**
- **Forward process**:
  - Image → gradually add noise → becomes random noise
- **Reverse process**:
  - Start from noise → remove noise step-by-step → image forms
- **Prompt** guides the generation at each step

